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Modeling die filling under gravity for different scales of rotary tablet presses

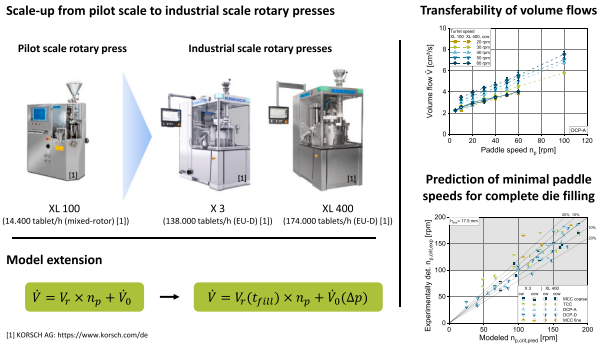
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HIGHLIGHTS

- Modeling the gravity filling process directly on industrial scale rotary presses based on a pilot scale model.
- Calculation of measured differential pressures based on Bernoulli's equation.
- Model extension to include the impact of evolving differential pressures on die filling.
- Prediction of the critical paddle speed based on few powder properties, machine and process parameters.

GRAPHICAL ABSTRACT



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ABSTRACT

For tableting, die filling is an important process step as it directly influences the tablet weight and its variation, which is connected to the drug dose. However, the impact of process parameters and materials properties is not sufficiently understood until today, so process development and scale-up of this process require a considerable amount of powder and time. Therefore, the aim of this study is to develop a process-model to describe and predict the filling process. The investigation aims a fundamental understanding of the die filling process under gravity with special regard to the scalability from pilot scale to industrial scale rotary presses using a feed frame with one paddle. In this context, the applicability of a recently developed model for gravity filling is investigated [1]. At larger scales an evolving differential pressure in the dies, facilitating the filling process, was identified, which was covered by a model extension based on Bernoulli's equation. Finally, the model allows the calculation of the critical paddle speeds required for complete die filling under gravity at certain production rates and for several pharmaceutical excipients based on a few powder properties, machine and process parameters.

1. Introduction

Die filling plays an important role in tablet production, which is the most commonly used oral dosage form in the pharmaceutical industry. The process on a rotary tablet presses can be divided into different sub-process steps: transport of powders to the feed frame, die filling,

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compaction and ejection. Die filling has a strong impact on critical product properties, especially on the uniformity of tablet weight and drug dose. Furthermore, inconsistencies of tablet weight result in high variations of the applied compaction stress which determines tablet porosity and tensile strength, which are important quality attributes as well [2,3]. Therefore, a stable and reliable die filling process is highly important in order to produce tablets of high quality and to minimize the number of rejected tablets.

The die filling process on a rotary tablet press is influenced by several factors. In general, die filling can be achieved by passive feeding (gravity feeding) by using a feed frame with static chambers as well as by active feeding, also called forced feeding, with rotating paddles inside a feed frame. In addition, the downward movement of the lower punch below the powder can create a suction pressure which enhances the powder flow into the dies. Furthermore, centrifugal forces arise as rotation frequencies of the turret and paddle are commonly high. Normal stresses resulting from powder above (e.g. in the filling pipe or only inside the feed frame), also referred to as overhead pressure, may lead to consolidation [3–8]. In the following, the impact of powder feeding with a paddle feeder and the effect of gravity and suction filling are discussed in more detail.

For die filling on rotary tablet presses, feed frames with rotating paddle wheels are commonly used, which can be distinguished by the number of paddle wheels. Usually, two-wheel feed frames are used in the production of single-layer tablets. One-wheel feed frames are used for the production of multi-layer tablets or on lab and pilot scale equipment, as the space for the feed frame on the pitch circle is reduced compared to single-layer production setups. In regard to the filled powder weight, especially the paddle speed has a considerable impact. Many studies found a favorable impact of an increasing paddle speed on the tablet weight and weight homogeneity [3,9–12]. For powders with poor flow properties, the influence is particularly pronounced [10,12]. Nevertheless, a high paddle speed can also have negative effects, as it enhances the dispersion of lubricants which may lead to overlubrication and may cause abrasion and breakage of particles [13–16]. Furthermore, high paddle speeds may result in densification of powders with good flow properties inside the dies [12]. Therefore, it is desired to use the minimal paddle speed which is necessary to fill the dies completely. On one hand, this value strongly depends on the powder properties – the worse the flow behavior, the higher the needed paddle speed [10,17] – and on the other hand on process parameters like dosing volume and turret speed. The higher the required delivery of powder into the dies per time, the higher the paddle speed has to be.

Two different filling mechanisms can be distinguished for die filling, gravity and suction filling [18]. During gravity filling, the lower punch is already in its low position (filling height) as the die moves below the feed frame. Here, filling takes place due to gravitational forces on the particles, while the escaping air flow counteracts the filling process [18]. Gravity filling is used if an external lubrication unit is applied or during multi-layer tableting from the second layer on, as the tamped first layer cannot move further downwards. During single-layer tableting and on the first layer of a multi-layer tablet, suction filling is commonly used to fill the dies. During suction filling, the lower punch is pulled downwards below the powder bed within the feed frame, resulting in a suction pressure as the inflowing air has to pass the powder [6]. Therefore, the suction pressure enhances the filling process by dragging the particles in dependency on the powder properties, effectively reducing the minimal paddle speed to completely fill the dies. As described above, an optimal process parameter selection has advantages, but is difficult to achieve as time and material-intensive experiments are necessary.

In the literature, a few researchers investigated the filling process on rotary tablet presses experimentally. Van Snick et al. evaluated the impact of powder blend properties of multiple formulations and process parameters on resulting tablet attributes by means of multivariate data analysis [3]. The developed PLS model can be used to predict tablet properties resulting from die filling. However, a general application of the PLS model remains questionable as the input parameters are not directly transferrable and the number values in the equation will accordingly change between different rotary presses. A deep insight into the impact of process parameters in combination with powder flowability is given by Peeters et al. and Mendez et al. who investigated die filling with two-paddle feed frames in a single-layer setup [10,17]. Yaginuma et al. studied the impact of turret speed on the tablet weight and its relative standard deviation for different grades of MCC and blends with lactose [9]. Here, for die filling passive feeding was used. Further studies investigated the design of feed frames and the shape of paddle wheels based on one excipient [19,20] as well as the type of fill cam for mini-tablet production [21]. A more fundamental work is done by Tang et al., who determined the filling ratio of different grades of microcrystalline cellulose on a model rotary die filling system with one paddle wheel under gravity [11]. They identified the turret speed as main impact factor on the filled weight while the effect also depends on powder properties, resulting in different critical turret speeds. However, results are mostly specific and hard to generalize for other materials or rotary presses.

In order to save material and time, some studies focused on simulations of powder flow into dies on rotary presses applying the discrete element method (DEM). Mainly, residence time distributions were studied, while feed frames with one up to three paddles were investigated [14,22–24]. Ketterhagen simulated die filling on the pilot scale rotary press XL 100 by using different paddle shapes and directions of rotation [22]. He found a higher mean tablet weight for a counter-clockwise rotating paddle wheel with an enlarged hub compared to clockwise direction, while the standard deviation depended strongly on the applied cohesion. However, it has to be noticed, that the feed frame possesses an extended volume on the left side which is not crossed by the paddle. Therefore, dead zones may develop there and it is unclear if the findings regarding the direction of rotation can be transferred to other feed frames with diverging designs. Mateo-Ortiz et al. investigated the number of spokes on the paddle wheel and their height on flow patterns, particle velocities and acting forces on the particles [25]. Their results present a back flow of powder above the paddles, depending on the distance between paddle and the lid. In the simulations, the waving behavior was only observed at high paddle speeds of 60 rpm. However, the powder wave behavior was also observed during experiments with a two-paddle feed frame at different paddle speeds [26]. In a simulative study, Mateo-Ortiz et al. found an empty space behind the spokes, which decreased with increasing number of spokes as the distance between them decreased. Consequently, a higher mean filling density inside the feed frame was found for a higher number of spokes, resulting in a reduced wave behavior and a more homogeneous powder bed height. This might be the reason for lower weight variability. The variability was slightly further decreased by using paddles with a height of 4.5 mm instead of 6.5 mm because the increased clearance between the paddle and the lid encourages the back flow, leading to an even more homogeneous powder distribution inside the feed frame.

In the context of digitization and quality-by-design approaches, it is desired to model single processes and combine them to process chain simulations. Flow sheet simulations or agent-based simulations incorporating (semi-)empirical or data driven models are used to investigate processes in-silico which is especially important for process optimization and dynamic process control in the context of continuous manufacturing [27–31]. Due to the high significance of modeling, it is important to consider sub process steps like die filling in the process chain of pharmaceutical tablets as well. On rotary tablet presses, studies mainly focus on the tablet weight variability, investigated with statistical methods and simulation approaches to identify important impact factors [10,32,33].

In order to enable the development of such process chain simulations and finally digital twins of processes with efficient and transferable process models, a mathematical model to describe the gravity filling process on a pilot scale rotary press with one paddle was recently

developed [1]. The model allows to predict the minimal paddle speed to completely fill the dies based on power properties and process parameters. The powder volume flow $\dot{V}$ into the dies was used for systematical analysis. It was calculated by dividing the mean tablet weight resulting from a parameter combination by the bulk density and the respective filling time. The filling time depends on the turret speed n_t and the filling angle α_{fill}:

$$t_{fill} = \frac{\alpha_{fill}}{n_t \times 360^\circ} \tag{1}$$

Following, a linear correlation between the volume flow $\dot{V}$ and the applied paddle speed n_p was found for not completely filled dies:

$$\dot{V} = V_r \times n_p + \dot{V}_0 \tag{2}$$

Here, the so-called rotation-specific fill volume V_r describes the change in volume flow rate over paddle speed (slope) and the so-called reference volume flow $\dot{V}_0$ scales the volume flow at a theoretical paddle speed of 0 rpm (axis intercept). Using this model, the critical paddle speed $n_{p,crit}$ to fill the dies completely can be calculated based on the dosing volume V_D, the turret speed n_t and the filling angle α_{fill}:

$$n_{p,crit} = \left(\frac{V_D n_t 360^\circ}{\alpha_{fill}} - \dot{V}_0\right) \times \frac{1}{V_r} \tag{3}$$

Reasonable procedures or models for scaling process parameters like the minimal paddle speed to fill the dies at desired turret speeds up to industrial rotary presses are not described until today. Therefore, the aim of this work is to investigate the scalability of this model onto rotary presses in production scale and further extend the model in order to reduce the time-to-market of new tablet formulations. Consequently, general understanding of influencing parameters is fostered and die filling under gravity with feed frames with one paddle wheel, as used for multi-layer tableting on the second layer or during external lubrication, can be described by this approach.

2. Materials and methods

2.1. Material

In this study, five pharmaceutical excipients with different solids density were used: anhydrous dicalcium phosphate (DCP-A, DI-CAFOS® A150, Chemische Fabrik Budenheim KG, Budenheim, Germany), dicalcium phosphate dihydrate (DCP-D, DI-CAFOS® D160, Chemische Fabrik Budenheim KG, Budenheim, Germany), tricalciumcitrate (TCC, Tricalcium Citrate, Jungbunzlauer, Ladenburg, Germany), microcrystalline cellulose in a coarse grade (MCC coarse, Vivapur® 200, JRS Pharma, Rosenberg, Germany) and a fine grade (MCC fine, Emcocel® 90 M, JRS Pharma, Rosenberg, Germany). DCP-A, DCP-D and TCC were mixed with magnesium stearate (LIGAMED® MF-2-V, Peter Greven GmbH & Co. KG, Bad Münstereifel, Germany) in a bin blender for 15 min at 30 rpm for lubrication. As seen in Fig. 1 a, the particle size distributions of the five different materials are close to each other. MCC coarse presents a wider particle size distribution with larger particles compared to the other materials, while MCC fine consists of smaller particles. Differences can be observed for the bulk, tapped and solids density (Fig. 1 b). The bulk density increases in the order MCC coarse, MCC fine, TCC, DCP-A, DCP-D, while DCP-A has the highest solids density.

2.2. Methods

2.2.1. Powder characterization

Particle sizes were measured by means of laser light diffraction (Mastersizer 3000, Malvern Panalytical, Malvern, United Kingdom). For particle dispersion, a dry dispersion unit AERO S was applied with a dispersion pressure of 0.5 bar. Three measurements were conducted and mean values were calculated. Bulk and tapped densities were measured according to the Ph. Eur. 9.3 2.9.34. A volumetric analyzer was used for tapping (Erich Tschacher Laboratoriumsbedarf, Bielefeld, Germany). Three measurements were conducted, respectively, and means were calculated. To calculate the Hausner ratio, the tapped density is divided by the bulk density of the respective material. For permeability measurements, a powder rheometer (MCR 302, Anton Paar, Graz, Austria) was used. The powder was preconditioned by fluidizing 80 ml powder at 5 l/min for 200 s. Simultaneously, an installed two-blade stirrer rotated with 5 rpm to ensure a homogeneous fluidization. After conditioning, an air-permeable punch was installed, which consolidated the powder bed by applying 1, 2 or 3 kPa. At each consolidation stress, volume flows of 0.5 and 1 l/min were applied and differential pressures were measured. Based on Darcy's law, the permeability was calculated [34,35]. The dynamic angle of repose was measured with the Granudrum (GranuTools, Awans, Belgium). 50 ml of loose powder was filled into the drum, which was sealed and rotated with frequencies of 2, 4 and 10 rpm. For each frequency, 20 images were taken. Subsequently, the analysis was conducted by segmenting the images and calculating the angle between the horizontal axis and the powder surface.

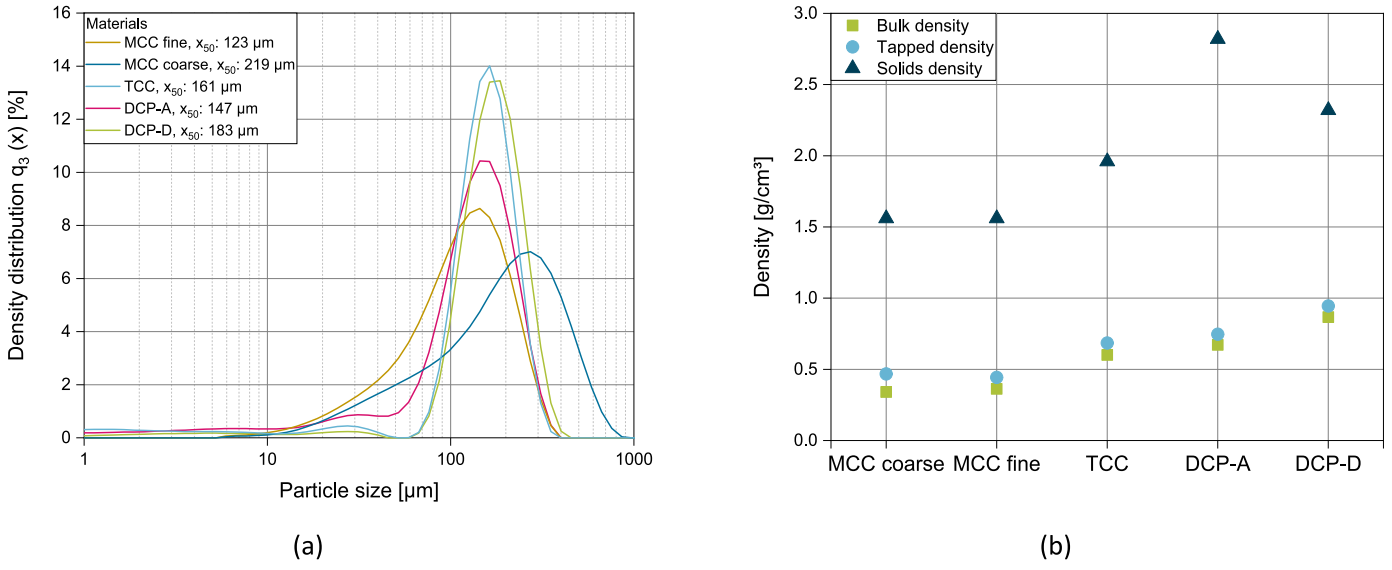


Fig. 1. Particle size distribution (a) and bulk, tapped and solids density of investigated materials (b).

2.2.2. Die filling experiments

In order to investigate the scale-up of die filling under gravity, one rotary press in pilot scale (XL 100, KORSCH AG, Berlin, Germany) and two rotary presses in production scale were used (X 3, XL 400, KORSCH AG, Berlin, Germany). The rotary presses mainly differ in their pitch diameters, which ranged between 118 mm (XL 100) and 410 mm (XL 400), as well as in the number of tools (Table 1). The XL 100 is equipped with a mixed rotor which possesses only four punch stations. The X 3 provides 23 tools, while the XL 400 has slightly more stations with 29 tools (Table 1). All rotary presses were equipped with round, flat-faced 9 mm EU-D tools.

On the XL 100, only feed frames with one paddle wheel can be installed. In order to enhance the comparability, feed frames with one paddle wheel were also used on the X 3 and XL 400. For the latter two, a two-layer setup was applied to investigate gravity filling. The lower punch is pulled down at the first filling station without having a feed frame equipped. In between the two filling stations, the lower punch remains in the lowest position. Subsequently, the dies enter the feed frame on the second filling station and are filled due to gravitational forces acting on the powder. Due to the two-layer setup, the turret speed on the X 3 and XL 400 may not exceed 60 rpm, which corresponds to the maximum turret speed of the XL 100 (Table 1). In contrast to the XL 100, where the maximum paddle speed is 60 rpm, the paddle speed on the X 3 and XL 400 can be increased up to 100 rpm.

On the XL 100, for model development in a previous study [1], four out of the five pharmaceutical excipients (all except MCC fine) were tableted with turret speeds of 20–60 rpm in steps of 10 rpm. After model development, MCC fine was additionally tableted to investigated the scalability. The turret rotated counter-clockwise, as for all investigated rotary presses. At each turret speed, the paddle speed was varied between 5 and 60 rpm (5, 10, 20, 30, 40, 50, 60 rpm). The paddle wheel exhibits 12 spokes with a height of 6 mm. The clearance between the paddle and the lid amounts to about 2.3 mm. During all experiments, the paddle wheel rotated clockwise. A fill cam with a filling height of 16 mm was used with a dosing height of 15.5 mm. Each die filling experiment was conducted three times.

For the trials on the X 3 and XL 400, turret speeds of 20–60 rpm in steps of 10 rpm were used, corresponding to the trials on the XL 100. Paddle speeds were varied between 10 and 60 rpm in steps of 10 rpm. A paddle speed of 100 rpm was investigated in addition. As there are no strict conventions on the direction of rotation, paddle wheels were rotated clockwise (cw) and counter-clockwise (ccw). The paddle wheel of the X 3 has 16 spokes with a height of 8 mm, while the wheel on the XL 400 possesses 12 spokes with a height of 6 mm, corresponding to the XL 100. The clearance between the paddle wheel and the lid differs as well for the industrial presses (X 3: 5.25 mm, XL 400: 2.5 mm). All paddle wheels have a rectangular and straight shape. For both industrial scale rotary presses, a fill cam with a filling height of 18 mm was used with a dosing height of 17.5 mm. For all experiments, filling and dosing height were kept constant. All experiments were carried out three times, while twenty tablets were collected and weighed with an automated tablet tester MultiCheck 6 (Erweka, Langen, Germany). Mean values and standard deviations were calculated for each process parameter combination. Following, volume flows into the dies were calculated for all three experiments and fitted together by using a linear equation.

2.2.3. Measurement of differential pressure

A differential pressure sensor (SDP810-500Pa, Sensirion, Stäfa, Switzerland) was used to measure differential pressures between the dies and the surrounding. The filling opening of an empty feed frame was sealed by tape in order to isolate the volume inside the filling opening between the tape and the turret against the environment (Fig. 2). A silicone tube was installed to connect the differential pressure sensor with the resulting air volume between the feed frame and the die table. The sensor possesses a measurement range of ±500 Pa against the ambient conditions with an accuracy of 3% of the measured value. For the pressure trails, only one lower punch was installed while all remaining dies were replaced by blind dies. The equipped feed frame was installed on the second layer to investigate the pressure changes on the station which exhibits gravity filling. The turret speeds ranged between 5 and 50 rpm and for each speed a measurement of 20–30 s with a

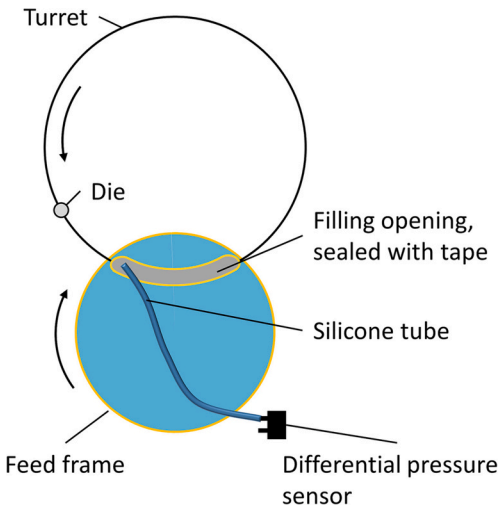


Fig. 2. Schematic illustration of the differential pressure measurement.

Table 1
Technical data of rotary presses XL 100, X 3 and XL 400 [36].





	XL 100	X 3	XL 400
Tools (EU-D)	4 (mixed rotor)	23	29
Pitch diameter [mm]	118	325	410
Turret speed (Bi-layer setup) [rpm]	20–60	5–60	5–60
Paddle speed [rpm]	20–60	5–100	5–100

measurement rate of 1000 per s. The resulting data was processed by using a moving average filter with 21 values. Following, mean values were calculated for each turret speed (comp. [1]).

3. Results and discussion

3.1. Die filling experiments

3.1.1. Volume flows on industrial scale rotary presses

In order to close the gap of scalability, the application of the gravity filling model on different scales of rotary presses is presented in this study. Furthermore, necessary model extensions to achieve a more precise description and prediction on larger scales were developed. Initially, the general applicability of the model has to be investigated by calculating the volume flow for the experiments on the two industrial scale rotary presses X 3 and XL 400 and check if they increase linearly with the paddle speed as well.

Fig. 3 presents the volume flows into the dies for DCP-A, obtained on the X 3 (Fig. 3 a, b) and XL 400 (Fig. 3 c, d) for clockwise and counter-clockwise rotation of the paddle wheel, respectively (graphs of other materials in Supplementary material Fig. S.1-S.4). For all four experiments, a linear course of the volume flow over the paddle speed, corresponding to the observations on the XL 100, can be found. Therefore, the previously developed linear model as described by Eq. (2) is generally applicable for the presented materials.

Besides the data for the industrial scale rotary presses, the graphs show volume flows obtained with the pilot scale rotary press XL 100 (Fig. 3). For the lowest turret speed of 20 rpm on the X 3 and XL 400 for certain constellations, the volume flows match the data collected with the XL 100. The overlap can be observed for DCP-A, DCP-D (except X 3 cw) and TCC on the XL 400 in ccw direction. For TCC in the other settings and MCC coarse and fine, volume flows on the industrial scale rotary presses at a turret speed of 20 rpm are below the volume flows observed on the XL 100. The reason for deviations may be based on the powder flow. Fluctuations in the flow into the feed frame through the filling pipe can be observed which indicates a lower flow than under optimal conditions. It can be assumed that these fluctuations also have an impact on the powder level inside the feed frame and thus affect the filling as well. Additionally, the number of spokes on the paddle wheel may have an impact. In simulation studies, a reduction of the empty

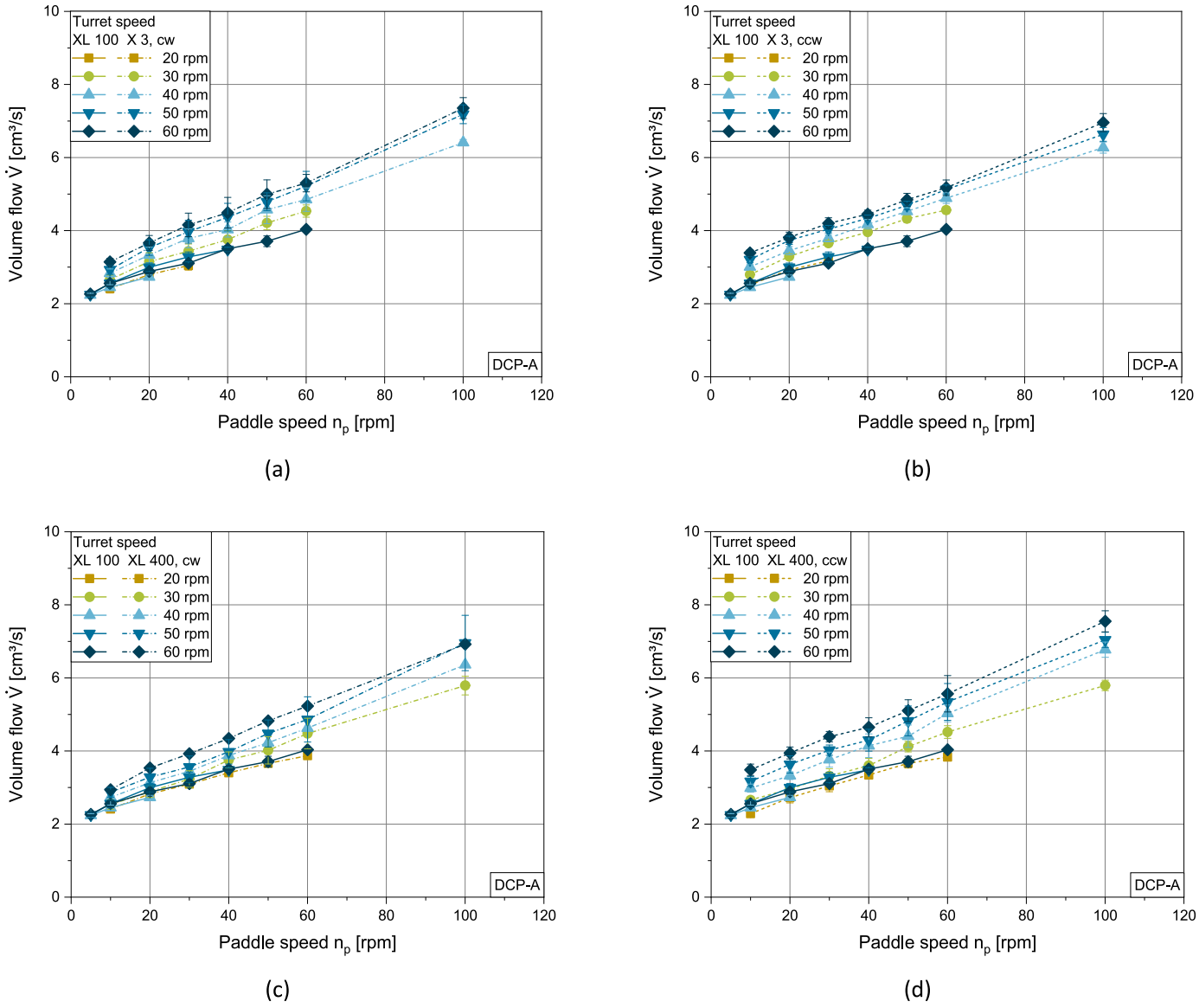


Fig. 3. Volume flow over paddle speed for DCP-A on the X 3 cw (a), the X 3 ccw (b), the XL 400 cw (c) and the XL 400 ccw (d). Representative results for one experimental repetition, respectively.

space behind the spokes due to back flow of powder was found for an increasing number of spokes, leading to a more homogeneous powder distribution and powder height inside the feed frame [25]. Since the X 3 possesses 16 spokes while the XL 400 has 12 spokes, this is a reasonable difference to explain the similar filling results in cw and ccw directions on the X 3 while for the XL 400 deviations between the directions can be observed. In addition, the X 3 has a higher clearance between the paddle and the lid which would allow a higher back flow compared to the XL 400. Consequently, the powder distribution inside the feed frame of the X 3 would be more homogeneous. For the studied setting and materials this may explain lower variations in the filling behavior for the different directions. The higher volume flow rate in ccw direction, found for the XL 400, correspond to the findings of Ketterhagen, who simulated die filling on a XL 100 [22]. However, for excellently flowing powders at low turret speeds, the volume flow based on Eq. (2) presents a directly scalable measure. With increasing turret speed, the volume flows split up and shift to higher values. While for cw directions an increasing slope is more pronounced, for ccw a higher parallel shift, resulting in a higher axis intercept, can be observed.

Overall, volume flows of three experimental repetitions (Fig. 3 displays $n = 1$) for each material and setting were plotted and fitted together for each turret speed by Eq. (2). For each material and setting an axis intercept called reference volume flow $\dot{V}_0$ and a slope called rotation-specific fill volume V_r are derived. For understanding the underlying reasons for the variations in volume flow rate with increasing turret speed, a deeper look on parameters which are connected to the turret speed must be taken. Furthermore, in the recent model, both parameters $\dot{V}_0$ and V_r are considered to be constant values, which results in the need of extending the model to describe the filling process on industrial scale rotary presses in detail and to predict optimal process parameters like the critical paddle speed to fill the dies completely.

3.1.2. Differential pressure measurement and estimation

In the literature, a considerable impact of the differential pressure, which changes with the turret speed, on the filled weight is presented [6,37,38]. Although no suction pressures are expected during gravity filling due to the missing downward motion of the punch, differential pressures were obtained in a previous study at the XL 100 [1]. Therefore, pressure measurements were conducted also on the XL 400. In general, differential pressure measurements present a baseline with fluctuations and discrete peaks at higher negative values in comparison to the baseline (Fig. 4 a). The baselines and discrete minima are shifted to higher negative values with increasing turret speed, while fluctuations rise simultaneously. The peaks at lower differential pressures correspond to the rotation frequency of the one only die, which is installed. As different side effects like possible turbulences during die passing, leakage between feed frame and turret as well as between the lower punch and the die walls may have an impact on the measurement, it is complicated to understand the cause of the peaks. For further analysis, mean values for the baseline are derived for the XL 400 and XL 100 (Fig. 4 b). The differential pressures on the XL 400 decreases with rising turret speed and reach considerably lower values compared to the XL 100 (Fig. 4 b).

Negative differential pressures may arise from the velocity of the air which is close to the turret and, thus, moves in a boundary layer as well as inside the dies. Based on these assumptions, the energy conservation based correlation between static pressure, dynamic pressure, which is linked to the velocity, and the fluid's potential energy based on Bernoulli can be applied [39,40]:

$$\frac{v_1^2\rho}{2} + \rho g h_1 + p_1 = \frac{v_2^2\rho}{2} + \rho g h_2 + p_2 \tag{4}$$

$$\Delta p = p_1 - p_2 = \frac{v_2^2\rho}{2} \tag{5}$$

In Eq. (4), state one (left to the equal sign) describes the energy level of the air around the turret, which is not moved ($v_1 = 0$), has a specific potential energy ($\rho g h_1$) and a static pressure (p_1). As the air is free to flow into the dies before entering the feed frame, this state has to have the same energy level in total as the air which moves inside the dies, described by state two (right to the equal sign). The velocity v_2 equals the circumferential speed of the turret and can be calculated based on the turret rotation frequency n_t and the radius r_t, on which the dies are distributed:

$$v_2 = 2\pi r_t n_t \tag{6}$$

$$\Delta p = \frac{(2\pi r_t n_t)^2 \rho}{2} \tag{7}$$

Following, state one and two are equal in potential energy which can therefore be neglected. Thus, the difference between the static pressures equals the dynamic pressure, based on the squared velocity v_2 and the fluid density ρ (Eq. (7)). The static pressure difference can also be measured as differential pressure against ambient conditions. The hypothesis that the mean differential pressure results from the rotation is supported by the good agreement between the calculated values and

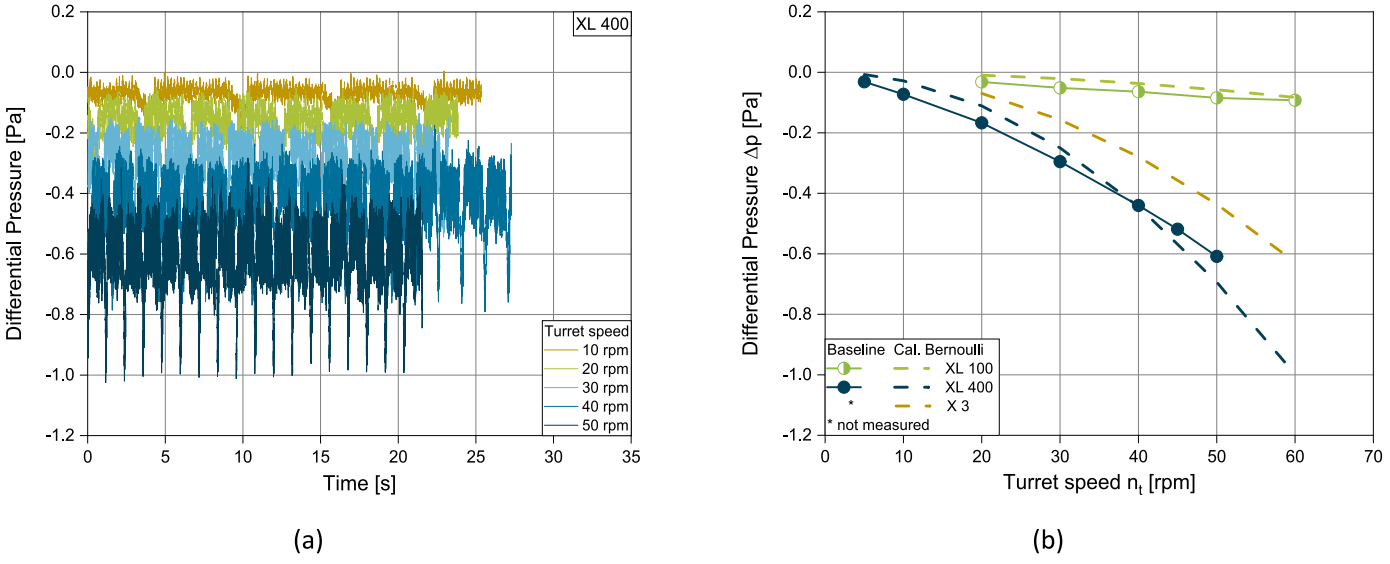


Fig. 4. Measured pressure curves (a) and derived mean and minimum differential pressures as well as pressures calculated by applying Bernoulli equation (b).

measured mean differential pressures on XL 100 and XL 400 (Fig. 4 b). As the turret radius is much smaller for the XL 100, circumferential speeds are lower compared to those on the XL 400 and consequently, much higher negative differential pressures are obtained for the XL 400. Based on these findings, it can be assumed that the differential pressures on the X 3 are less negative than the ones of the XL 400, but close to them (Fig. 4 b). Consequently, also during gravity filling, a differential pressure arises in the dies. However, this negative pressure is assumed to be smaller compared to values resulting if the lower punch is pulled down below the filling opening. Nevertheless, the rising differential pressure may enhance the flow into the dies with increasing turret speed, which is a the most probable explanation for the shift of volume flows to higher values.

3.1.3. Approach to model the impact on the reference volume flow $\dot{V}_{0,app}$

In case of industrial scale rotary presses, the reference volume flow $\dot{V}_0$ depends on the turret speed, as presented in Fig. 3. However, this correlation was not found for the XL 100 during previous investigations [1]. For a clear differentiation from the reference volume flow developed on the XL 100, the process parameter dependent variable is called apparent reference volume flow $\dot{V}_{0,app}$. Since the model is intended to be applicable to all scales, the turret speed itself can therefore not be used as a scaling parameter for the shift. Therefore, an indirect process parameter like the occurring differential pressure may be the reason for the shift and, therefore, a good scaling parameter. By plotting the respective parameters, correlations between the calculated differential pressures Δp (Eq. (7)) and the reference volume flow $\dot{V}_{0,app}$ on the X 3 and XL 400 can be identified (Fig. 5). For all investigated materials, an increase in $\dot{V}_{0,app}$ with rising Δp can be observed. In general, considerable differences between the materials are apparent. TCC, MCC coarse and MCC fine show similar values for cw and ccw directions of rotation, while DCP-A and DCP-D show higher reference volume flows in ccw direction for both rotary presses. On the XL 400 ccw, MCC coarse exhibits considerably higher $\dot{V}_{0,app}$ with high standard deviations, resulting from differences in three experimental repetitions (Supplementary material Fig. S.5). The high fluctuation may be caused by the medium flow properties and high potential of consolidation regarding the Hausner ratio of the material, while all other materials present a good reproducibility.

Reference volume flows from the XL 100 are plotted at differential pressures of 0 Pa, as $\dot{V}_0$ was not considered to change over XL 100 turret speeds [1]. Additionally, mean differential pressures are small. It can be emphasized that $\dot{V}_{0,app}$ of the X 3 and XL 400 match well with the $\dot{V}_0$ of the XL 100 in linear extension. As it is beneficial to use the model for a pilot scale rotary press as a basis to model larger presses, the data of the X 3 and XL 400 are fitted linearly for cw and ccw, respectively.

Following, the respective axis intercepts $\dot{V}_{0,app}(\Delta p = 0)$ and the slopes of the fits are investigated and correlated with measured powder properties. For the model describing a pilot scale rotary press, a correlation between the $\dot{V}_0$ and the bulk density ρ_b was identified [1]. In case of different machine scales, a plot of the axis intercepts $\dot{V}_{0,app}(\Delta p = 0)$ over ρ_b, including the data found for the XL 100 under gravity conditions, shows relatively good agreement as well (Fig. 6 a). Additionally, no clear differences between the rotary presses and the direction of rotation of the paddle wheel can be identified. Consequently, the developed model equation based on the XL 100 is applied to predict the $\dot{V}_{0,app}(\Delta p = 0)$ for all materials in dependency on their bulk density (Fig. 6 a) [1]:

$$\dot{V}_{0,app}(\Delta p = 0) = 2.26\,\rho_b + 0.42 \tag{8}$$

Regarding the change of the reference volume flow with the differential pressure, a rough correlation with the powder permeability K at a consolidation stress of 1 kPa was found (Fig. 6 b). Strong deviations between the different settings as well as for each material itself are apparent. In general, the X 3 exhibits higher negative values compared to the XL 400, corresponding to a higher change of reference volume flow with differential pressure. For the XL 400, slopes for cw rotation are almost zero for materials with even high permeability, which does not follow the general trend. Possible impact factors, like the number of spokes on the paddle wheel or general geometric specificities of the feed frames like non-symmetrical designs, may have an effect on the change in $\dot{V}_{0,app}$ for cw and ccw direction with differential pressure. Differences in the space above the paddles inside the feed frame for the X 3 and XL 400 may also contribute to variations, as discussed above. Nevertheless, these effects are not clearly identifiable based on the present data, so further studies regarding the elucidation of such impact factors should be conducted in the future. Therefore, a linear fit, taking all data points into account and passing through the origin, is applied. Finally, the reference volume flow's dependency on the differential pressure can be calculated as follows:

$$\dot{V}_{0,app}(\Delta p) = -1E11K \times (2\pi r_t n_t)^2 \frac{\rho}{2} + 2.26\,\rho_b + 0.42 \tag{9}$$

Differences between each of the fits and the experimental values will cause errors in the prediction of minimal paddle speeds. A separate

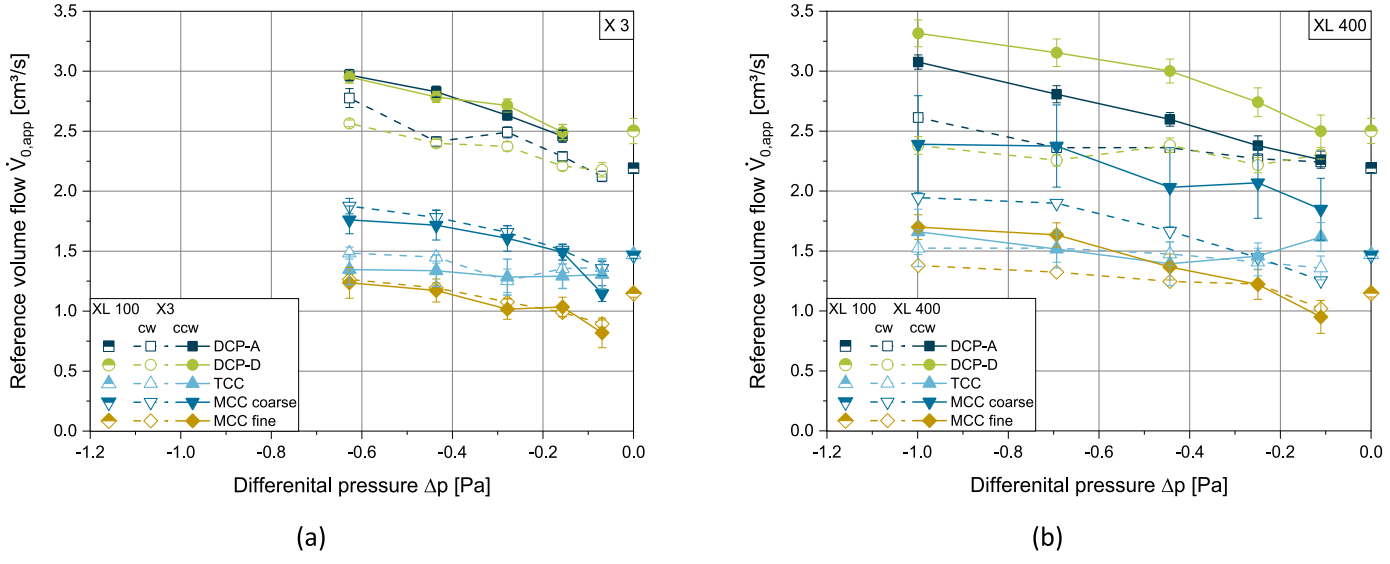


Fig. 5. Reference volume flows $\dot{V}_{0,app}$ for all investigated materials on the X 3 (a) and XL 400 (b).

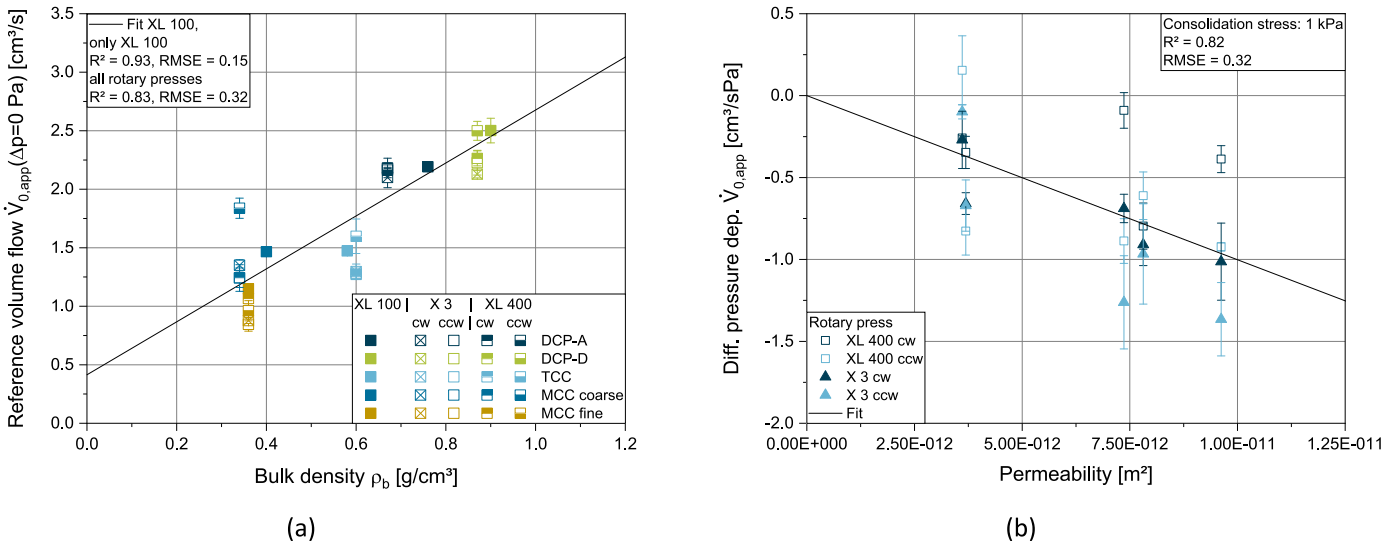


Fig. 6. Correlation between the axis intercept of reference volume flow $\dot{V}_{0,app}(\Delta p = 0)$ and the bulk density for all investigated rotary presses (a); change of the reference volume flow $\dot{V}_{0,app}$ with differential pressure over permeability (b).

model equation for each scale or direction of rotation, without identifying the specific determinants for the scale would lower potential errors but simultaneously it would drastically reduce the probability of a general applicability to other rotary presses. However, due to the aim to develop a scale-up model which covers various rotary presses, is simple and requires only a minimum of input data like powder and machine properties, this simplification and the corresponding errors are accepted. Deviations are discussed in the following.

The predictive strength of this equation can be checked by calculating the reference volume flows $\dot{V}_{0,app}$ for the maximum turret speed of 60 rpm and therefore the highest differential pressures. For most materials and settings, calculated and experimental data points are close to each other (Fig. 7 a). It can be observed that the experimental reference volume flows differ for cw and ccw, while the calculated values are independent of the direction of rotation which leads to deviations depending on the material. For TCC and DCP-D (cw, X 3 and XL 400), experimental values are considerably lower than the calculated values. Reference volume flows of TCC are overestimated by 30–50% over all turret speeds, while the maximum deviation for DCP-D ranges between 10 and 30% for the cw direction of rotation (Supplementary material Fig. S.6 b, c). For MCC fine, high deviations of about 20–50% can be found for a turret speed of 20 rpm, decreasing to below 20% with rising turret speed. Relative errors for all investigated materials can be found in the supplementary material (Supplementary material Fig. S.6). For MCC coarse, an increased reference volume flow with high standard deviation was found on the XL 400 in ccw direction, resulting from strong deviations between the three experiments.

In order to have a closer look at the trend of the reference volume flow over the turret speed, experimentally determined and calculated values are presented exemplarily for DCP-A (Fig. 7 b). In general, a good agreement can be found for this material. Especially values for cw direction are matching well. However, the experimental data reveals differences for a ccw rotating paddle as the direct of rotation is currently not considered in the model as a systematic correlation hast not been found, yet. Calculated values only distinguish between the two rotary presses as the expected differential pressures are different. Consequently, further impact factors have to be identified and included in a model to describe machine-specific effects in direction of paddle

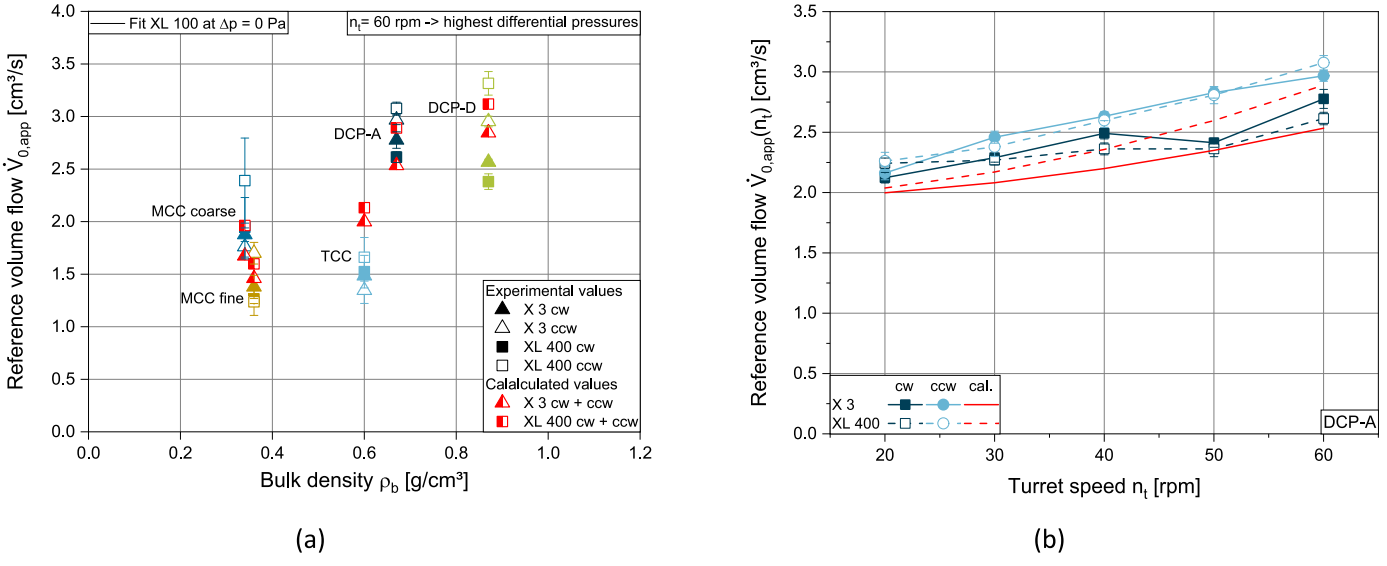


Fig. 7. Experimentally determined and calculated reference volume flows at a turret speed of 60 rpm (a); reference volume flow over turret speed for DCP-A (b).

rotation. A possible reason for the differences could be variations in the flow pattern of the powder inside the feed frame and the filling area. As the powder enters the filling area, pushed by a clockwise rotating paddle, it is partially moved with the turret to the end of the filling area, where the spokes remove it. For a ccw rotating paddle, powder and turret move in opposite directions in their area of contact, which may result in a higher amount of powder which is effectively present in the filling area for die filling. For this implementation, future studies should more closely investigate rotation direction-wise specificities in the feed frame geometries and deduce terms for filling models.

Systematic deviations of the reference volume flow will have an impact on the predicted critical paddle speed. An overestimation leads to predicted critical paddle speeds which are lower than the experimentally obtained values and vice versa. However, the aim of this study is to develop a general approach to model die filling between different scales which copes with different materials with varying properties, resulting in individual errors as a consequence. The deviations of the predicted paddle speed are presented in discussed in section 3.1.5.

3.1.4. Approach to model the impact on the rotation-specific fill volume V_r

The volume flow rates over paddle speed do not only shift parallel to higher values with increasing turret speed but also increase in their slopes, called rotation-specific fill volume V_r (Fig. 3). To include this in the model, a correlation with the turret speed was considered. However, a change over turret speed would be in conflict with the approach of a constant rotation-specific fill volume for the pilot scale rotary press. In order to take the machine scale into account, the filling time based on the filling angle was calculated (Eq. (1)). A plot of the rotation-specific fill volume V_r over the filling time presents a good correlation and, thus, is fitted for the X 3 and XL 400 in both directions of rotation (Fig. 8, exemplarily showing DCP-A). Here, a reciprocal function $1/x$ is applied, as the rotation-specific fill volume rises for declining filling times:

$$V_r\left(t_{fill}\right) = a \times t_{fill}^{-1} + b \tag{10}$$

Here, parameter a describes the curvature over the filling time, while parameter b is a measure for the offset to a limit value of the rotation-specific fill volume at high filling times. For DCP-A and DCP-D, V_r derived from the XL 100 matches the overall trend very well (Fig. 8, Supplementary material Fig. S.7 a). However, this cannot be observed for all investigated materials, as the rotation-specific fill volume is higher on the pilot scale rotary press for TCC and both grades of MCC (Supplementary material Fig. S.7 b-d). For cw direction, data points for both rotary presses are close to each other and the resulting fit function. For all materials, V_r is higher in ccw direction on the XL 400, while, in contrast, it is lower for ccw on the X 3 (comp. Supplementary material Fig. S.7). Again, differences in feed frame and paddle design or variations in the flow pattern in filling pipes and feed frames in dependency on powder properties may have an impact on these trends.

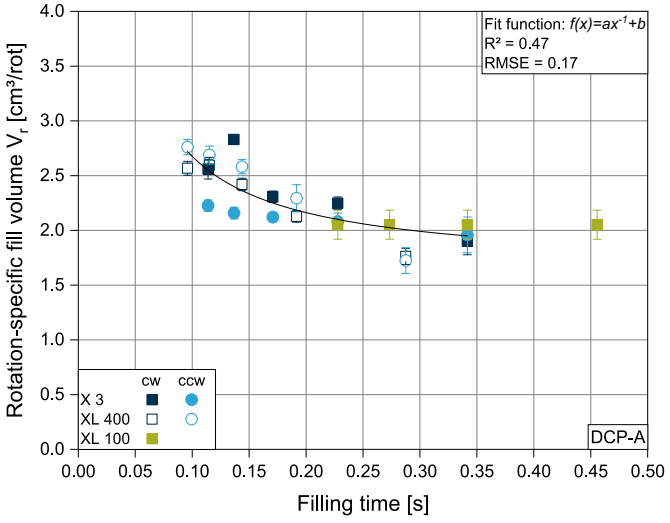


Fig. 8. Rotation-specific fill volume over filling time for DCP-A.

Corresponding to the model development for the reference volume flow $\dot{V}_{0,app}$, correlations of the rotation-specific fill volume V_r with powder properties are studied. For the XL 100, a negative proportionality with the permeability was found. However, a correlation with this powder property was not sufficient for the industrial scale presses (supplementary material fig. S.8), which emphasizes the complexity of impact factors and effects, which needs to be understood in more depth. Nevertheless, investigations of the fit parameters factor a and the offset b for the investigated materials present good linear correlations for a with the Hausner ratio HR (Fig. 9 a) and for b with the dynamic angle of repose α_{dyn}, measured at a rotation frequency of 10 rpm (Fig. 9 b).

Both correlations are generally reasonable as they describe compressibility and flow behavior under dynamic conditions comparable to the powder stressing in the feed frame. The Hausner ratio HR describes the relation between the tapped density and the bulk density and, thus, is a measure for the compressibility of the powder under gravitational forces. It is generally known that compressible powders are challenging to handle, e.g. if it comes to caking during storage or bridging inside a funnel or pipe. Furthermore, compressible powders often exhibit poor flow properties. For compressible powders with high Hausner ratios, the rotation-specific fill volume remains almost constant over the filling time, while for poorly compressible powders, an increase can be observed:

$$a = -0.52HR + 0.70 \tag{11}$$

The shift of the rotation-specific fill volume to higher values (offset b) correlates with the dynamic angle of repose α_{dyn} (Fig. 9 b):

$$b = -0.04\alpha_{dyn} + 3.05 \tag{12}$$

The lower the dynamic angle, and thus the better the flowability, the higher the baseline of the rotation-specific fill volume V_r over filling time. This means a higher volume flow per rotation of the paddle can be provided for die filling. Here, a connection with the dynamic angle of repose makes sense, because with lower angles, more powder can flow into the die from the neighboring area. Therefore, more powder is available for die filling at the same paddle speed.

Combining Eq. (11) and Eq. (12), the rotation-specific fill volume V_r can be described as follows:

$$V_r\left(t_{fill}\right) = \left(-0.52HR + 0.70\right) \times t_{fill}^{-1} - 0.04\alpha_{dyn} + 3.05 \tag{13}$$

The predicted values for $V_r(t_{fill})$ match the obtained values based on the experiments well, which can be exemplarily seen for DCP-A (Fig. 10). However, values for ccw direction on the X 3 are overestimated for smaller filling times, while on the XL 400 they are slightly underestimated. The trend of a higher predictive accuracy for cw direction can also be found for the other materials (Supplementary material Fig. S.9). In general, relative errors are considerably small for all materials for cw rotation compared to ccw. As stated for DCP-A, also for the other materials contrasting trends for the over−/underestimation of V_r are apparent for the X 3 and XL 400 (ccw). The different number of spokes per paddle (X 3: 16, XL 400: 12) may have an impact here. However, it is surprising that the diverging impact is only apparent for one direction of rotation.

3.1.5. Predictive power of the model

Finally, after taking the impact on the reference volume flow and the rotation-specific fill volume based on powder properties into account, the complete equation to describe the volume flow into the dies can be set up by combining Eqs. (2), (9) and (13):

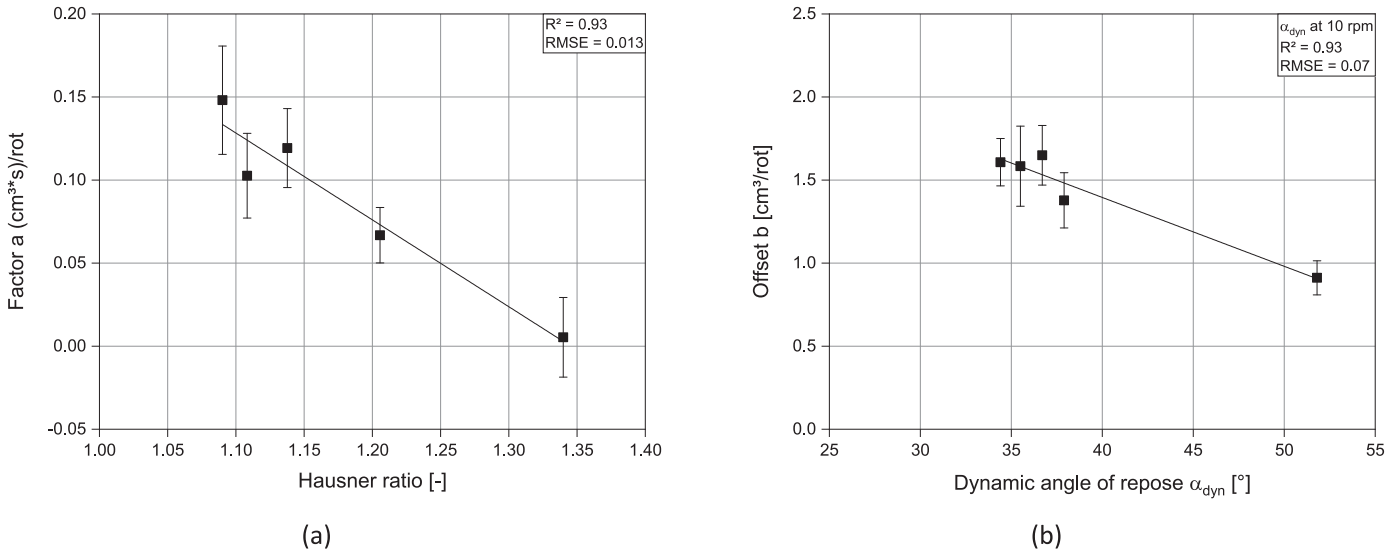


Fig. 9. Correlation between factor a (a) and offset b (b) of the rotation-specific fill volume V_r over filling time and the Hausner ratio and the dynamic angle of repose, respectively.

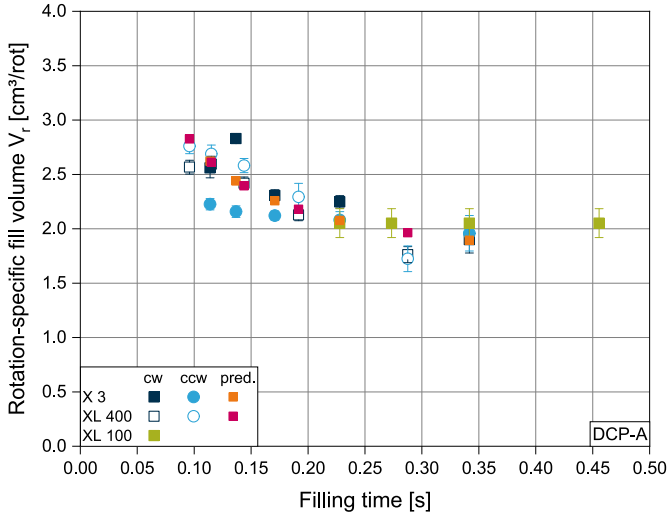


Fig. 10. Trend of the rotation-specific fill volume $V_r(t_{fill})$ for DCP-A over the filling time, experimentally obtained at the X 3 and XL 400 and calculated based on Eq. (13).

$$\dot{V} = \left(\left(-0.52HR + 0.70 \right) \times t_{fill}^{-1} - 0.04\alpha_{dyn} + 3.05 \right) \times n_p + \left(-1E11K \times \left(2\pi r_t n_t \right)^2 \frac{\rho}{2} + 2.26\rho_b + 0.42 \right) \tag{14}$$

In this model, only independently measured powder properties are used in combination with empirically determined correlations to predict the volume flow into the dies over three machines scales. In order to proof the applicability and predictive strength of the model, the critical paddle speeds $n_{p,crit}$ to fill the dies completely were calculated (based on Eq. (3)) and compared with the values determined in the experiments. Here, volume flow rates of three experiments were fitted for each turret speed, so reference volume flow $\dot{V}_{0,app}$ and rotation-specific fill volume V_r were experimentally obtained. With these values, Eq. (3) was applied to calculate the critical paddle speeds $n_{p,crit}$. Relative and mean relative errors for each material and rotary press with the respective direction of rotation were calculated in relation to the experimentally obtained values (Fig. 11). Values lower than 0 correspond to an underestimation of the critical paddle speed by the model and vice versa. Relative errors present a good predictive strength with errors around ±25%. Only at a low turret speed of 20 rpm, generally higher relative errors can be observed (Fig. 11 a). This can be traced back to fits, which are based on only a few data points, especially on the X 3, as dies are already completely filled at low paddle speeds. Furthermore, MCC coarse presents high relative errors for the experiments conducted on the XL 400 in ccw direction. Deviations for this setting were already observable during model development. For this setting, deviations of the volume flow between the three experiments are high (Supplementary material Fig. S.5), which may be due to fluctuations in powder flow through the filling pipe and inside the feed frame. As MCC coarse presents the highest Hausner ratio, it is most prone to consolidation which affects the powder flow. Additional impact factors are environmental conditions, as the flowability and electro static charging of the MCC particles depend on the relative humidity, which ranged between 20% and 40%. The higher deviations for MCC are also reflected in the elevated mean error over all turret speeds (Fig. 11 b). Generally, based on the mean errors, it can be stated that the die filling is described well for both rotary presses in cw direction, while errors for ccw direction can be found above (XL 400) and below (X 3) zero, which indicates an overestimation of the critical paddle speed on the XL 400 and an underestimation on the X 3. This finding corresponds to the systematic errors found for the rotation-specific fill volume (comp. 3.1.4 and Supplementary material Fig. S.9).

The good predictive power is further emphasized by the parity plots for dosing heights of 17.5 mm in cw and ccw diretion (Fig. 12). In this study, a challenging setup with a dosing height of 17.5 mm was used, resulting in the maximal adjustable paddle speed of 100 rpm of the rotary presses X 3 and XL 400 being greatly exceeded for most turret speeds and materials. However, the deviation is about ±25%, which is acceptable for estimating the minimal paddle speed based on a model for a pilot scale rotary press with extensions based on the permeability, the Hausner ratio and the dynamic angle of repose. It is remarkable that this relatively simple model covers a wide range of settings and still results in precise to acceptable values for the critical paddle speed. Especially, if the relative errors for $\dot{V}_{0,app}$ and V_r, discussed above, are considered. In order to further improve the model, additional impact factors have to be identified and studied separately. As discussed, the number of spokes on a paddle wheel could have an impact as well as the flow pattern inside the feed frame. Furthermore, also the actual fill level of the feed frame could affect the results. Especially for the different directions of rotation, a rational explanation for the differences in the filling process are

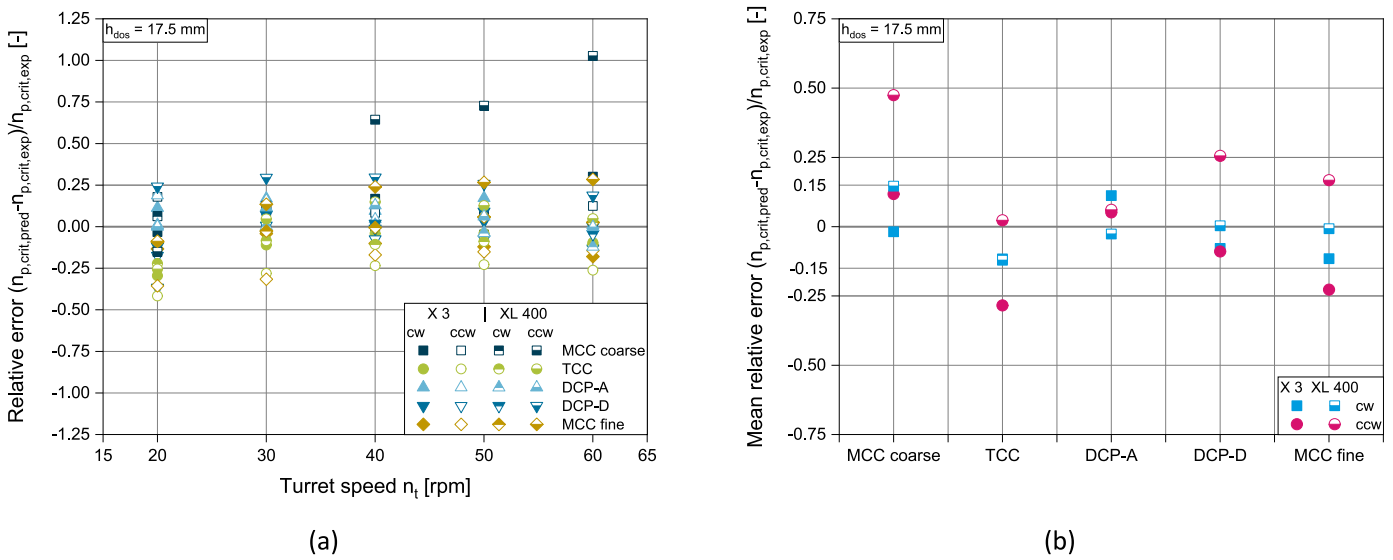


Fig. 11. Relative error of the predicted critical paddle speed in relation to value determined in experiments over turret speed (a); mean relative error of the predicted critical paddle speed for the investigated materials (b).

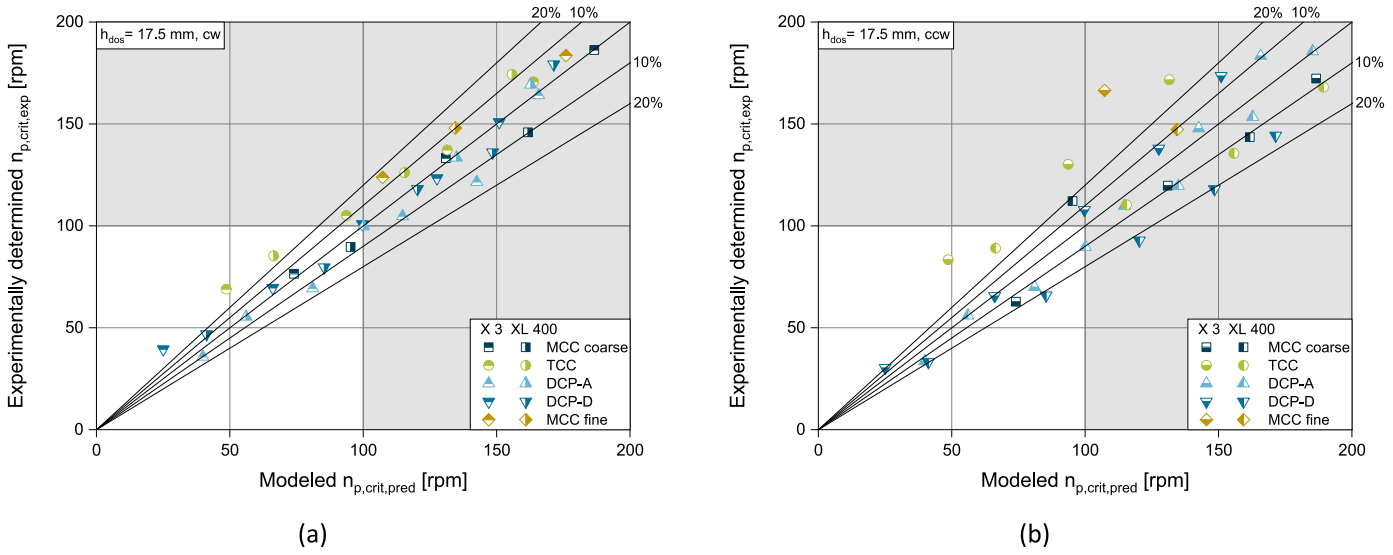


Fig. 12. Parity plot for the modeled critical paddle speed $n_{p,crit,pred}$ and the experimentally determined critical paddle speed $n_{p,crit,exp}$ for a dosing height of 17.5 mm in (a) cw and (b) ccw direction. Since paddle speeds of up to 100 rpm can be set for the rotary presses used, areas with theoretically higher paddle speed are marked in gray and serve for demonstration.

necessary to improve the predictability of the minimal paddle speed. For application, the critical paddle speed determined by the model equation can be multiplied by a safety factor of 1.25 to gain a start value for further process design. Therefore, the developed model provides a very good basis for an ad initio determination of critical process parameter settings of the die filling process over machine scales from lab to production. Additionally, these predicted values can be optimized by pinpointed experimental investigations.

4. Conclusion

In this study, a model for die filling on a pilot scale rotary tablet press was extended to allow the scale-up to tablet presses in industrial scale. The model can be used to predict the critical paddle speed to completely fill the dies on industrial scale rotary presses with one paddle under gravity filling based on only a few powder properties (bulk density, permeability, Hausner ratio and dynamic angle of repose) which are readily accessible with only small quantities of powder. For volume flow rates of excellently flowing powders, an overlap was found between values obtained for pilot scale and a low turret speed of 20 rpm on industrial scale presses. With increasing turret speed on the industrial presses, volume flow rates shift to higher values by exhibiting higher axis intercepts (reference volume flow $\dot{V}_{0,app}$) and rising slopes (rotation-specific fill volume V_r). These findings require a model extension for the parameters. Here, measured differential pressures over the moving dies, which can be estimated by the static pressure reduction calculated by Bernoulli's equation, correlate with the increase of the reference volume flow $\dot{V}_{0,app}$ with regard to the permeability of the respective powder. The rotation-specific fill volume V_r was found to increase with decreasing filling time with fit parameters depending on the Hausner ratio and the dynamic angle of repose. By combining the correlations for the reference volume flow and the rotation-specific fill volume, a linear equation to describe the volume flow for the five investigated materials on the differently scaled rotary presses was set up.

The predictive strength of the model was investigated by comparing the predicted and experimentally obtained critical paddle speeds to fill the dies completely. In general, a good prediction was achieved with relative errors of mostly smaller than 25%, which generally enables a scale-up of die filling based on a model basis for the first time. This approach accelerates the process development for the industrial scale rotary presses and reduces its material consumption. Overall, it can result in more stable processes and tablets with higher quality as the risks of overlubrication and particle breakage in the filling system are reduced by optimizing the paddle speed.

The final model, calibrated on the investigated powders, only takes independently measured powder properties and manipulable machine parameters into account. In case of applying it to other rotary presses and materials, it can be easily adapted. In the future, it will be investigated how well the presented correlations can be applied to other materials. Additionally, an extension of the model to incorporate feed frames with two paddles as well as suction filling will be investigated.

CRediT authorship contribution statement

Ann Kathrin Schomberg: Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Arno Kwade:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing. **Jan Henrik Finke:** Conceptualization, Investigation, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by Korsch AG. Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by Chemische Fabrik Budenheim KG. Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by Jungbunzlauer Ladenburg GmbH. Ann Kathrin Schomberg reports equipment, drugs, or supplies was provided by J Rettenmaier and Sons.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.powtec.2023.118550.

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